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Course Name: Computer Networks Course Code: CSA0702

1) Application → Presentation → Session → Transport →

Network → Data Link → Physical

Missing OSI Layers and their Functions

- Session Layer - Layer 5:

Essential for managing, maintaing, and terminating connections between applications. It enables reliable communication by ensuring that data streams are kept separate and can be synchronized. Without session layer, your app wouldn't know if it's talking to the right process.
Ex: You open Netflix + Chrome at same time. Session layer makes sure Netflix data goes to the right app (Netflix), not chrome.

- Data Link Layer - Layer 2

Responsible for node-to-node delivery and framing. It ensures reliable communication by providing flow control and error control at the physical medium level, ensuring that bits are correctly framed and transformed and transmitted between adjacent devices. Network layer only gets data from IP to IP. But Data Link makes sure it actually reaches the correct NIC card without errors.

2) Concept Map : Data units and OSI Layers

The relationship between layers and data units is :

- Transport layer : Segment
- Network layer : Packet
- Data Link layer : Frame
- Physical layer : Bits

Data transformation - encapsulation - occurs as data moves down the stack. This process is called Encapsulation on Sender side and De-encapsulation on receiver side.

Sender sid : Top to Bottom Encapsulation

1) Transport → Segment

Data from app is broken into segments. TCP/UDP header added with Port numbers. Port numbers tell which app should receive it. Also handles flow control.

2) Network → Packet

Segment gets IP header added. Now it's a Packet. IP address tells which device on the internet to send to. Enables routing across networks.

3) Data Link → Frame

Packets gets MAC header + trailer added. Now it's a Frame.

4) Physical → Bits

Frame is converted to 0's and 1's and sent as electrical / optical / radio signals.

3) Explanation: Datalink + Network Layers for complete delivery

1) Network Layer ⟶ IP Address ⟶ Gilobal Delivery

Operates across interconnected Networks. The IP address acts as a logical identifier, enabling routers to determine the path (routing) required for a packet to traverse multiple networks to reach a destination globally. Get the data from one network to another network, anywhere in the world. The IP address are logical and can work across different networks. The Router looks at the Destination IP and decides: "Sond this to the next router towards USA". IP gets the packet to the correct network/city/county. But it doesn't know which exact device in that house.

2) DataLink Layer ⟶ MAC address → Local Delivery

Get the data from one device to another device within the same local network. Operates on the physical network segment. The MAC address uniquely identifies devices on the same local network, allowing for reliable delivery between adjacent nodes. MAC gets the frame to the exact device in the room. But MAC's don't work across routers.

4) Error Control: DataLink Layer + Transport Layer

Error correction/control happens in two stage.

1) Datalink Layer (Error Detection):

Focuses on detecting transmission error at the physical link level (e.g., using checksums or CRC). It ensures that frames arriving at the next hop are intact. Only detects error. It cannot fix them. Drops the bad frames and asks sender to resend via ARQ - Automatic Repeat Request.
Example: Wifi frame gets noise. CRC fails → Frame dropped.

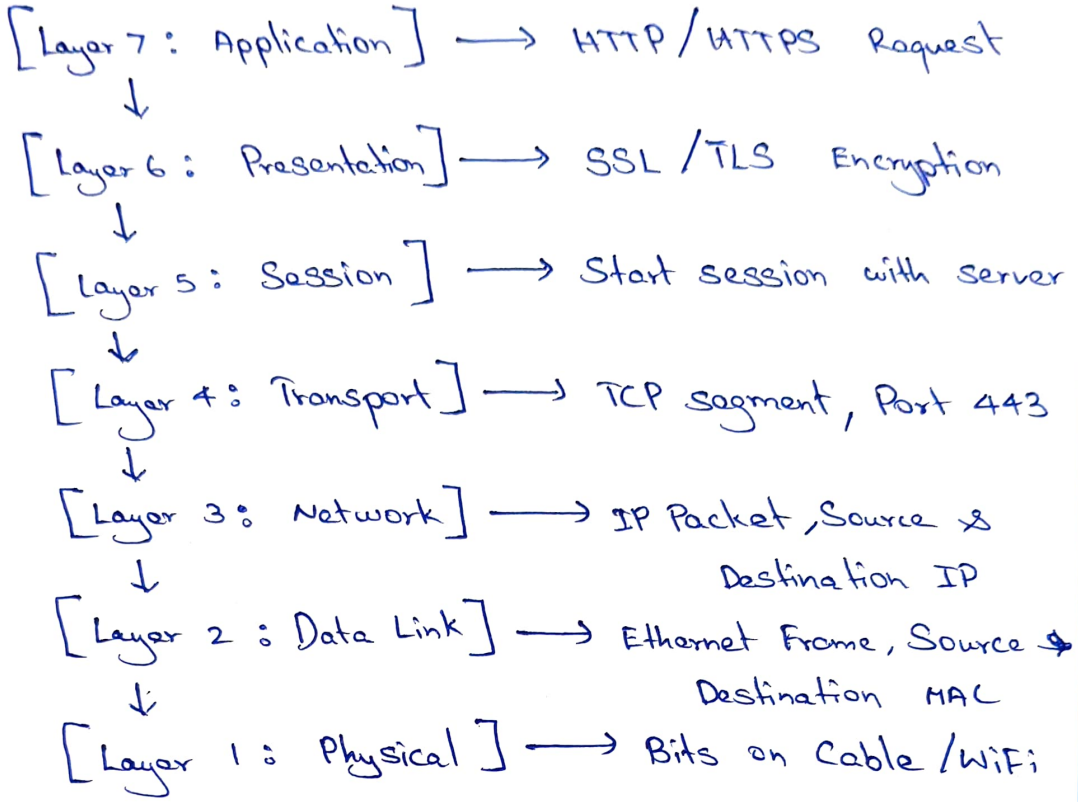
2) Transport Layer (Error Correction):

Ensure end-to-end reliable delivery between 2 applications. Ex: Your chrome to Google Server

- Segmentation: Data broken into segments + Sequance numbers added.
- Acknowledgement: Receiver sends ACK for each segment received.
- Retransmission: If sender doesn't get ACK in time, it resends the segment.
- Flow control: Prevents overwhelming.

Corrects errors by resending lost/corrupted segments.

5) Concept Map: Web Browsing using OSI Layers



Failure Analysis:

If a lower layer fails (e.g., physical layer cable disconnect), the entire chain is broken, leading to complete loss of connectivity. If a higher layer fails (e.g., Presentation), the entire chain is broken, leading to a complete loss of data, the underlying transport will remain active, but the application will be unable to interpret the data, resulting in a failure to display the webpage. Because the OSI model is hierarchial, a failure at any stage prevents the subsequent processing required for successful web browsing. Follow-ups.